AIDL

Unit - 1

1) Explain the problem-solving Agents with an example.

problem-solving agent is a goal-based environment agent that decides what sequence of actions to take in order to reach a desired goal from an initial state.

→ It does this by formulating a problem and searching for a Solution

→ Components of a problem-solving agent.

1. Initial state
2. Actions (operators)
3. State space
4. Goal state
5. Path Cost

Working of a problem-solving Agent.

1. Perceive the environment.
2. formulate a goal
3. formulate the problem
4. Search for a solution
5. Execute the solution.

Diagram :-

Environment

↓

percepts

↓

problem-solving Agent

↓

Actions

↓

Environment

Example :-

8-puzzle problem.

1. Initial state: scrambled tiles
2. Goal state: ordered tiles
3. Actions: move blank tile up/down/left/right
4. Solution: sequence ~~of~~ tiles moves.

2. Explain the nature of Environments in detail.

The nature of an environment describes the characteristics of the world in which an intelligent agent operates. It helps in deciding which type of agent is suitable for a task.

properties of Environment :-

1. Fully Observable vs partially Observable.
2. Deterministic vs problematic (stochastic)
3. Episodic vs Sequential
4. Static vs Dynamic
5. Discrete vs Continous
6. Single Agent vs Multi-Agent.

1. Fully observable Environment :-
   - Agent's sensors give complete information about the environment
   - No hidden states.

   Example : chess game.

2. Deterministic Environment :-
   - Next state is completely determined by current state and action.

   Example: puzzle solving.

3. Episodic vs sequential :-
   - Each action is independent of previous actions.

   Example: Image classification.

4. static vs Dynamic
   - The environment doesnot change while the agent is deciding

   Example :-
   - Crossword puzzle

1. Initial state : scrambled tiles

5. Discrete vs Continuous

- Finite number of states, actions or percepts

Example :

- Chess moves.

6. Single Agent vs Multi - agent

- Only one agent operates

Example : crossword puzzles

3. Explain the Classical planning problem.

- A classical planning problem in AI is one in which an agent, operating in a fully observable, deterministic, static and discrete environment, finds a sequence of actions that transforms an initial state in a goal state.

Core Assumptions :-

1) Fully observable - Complete knowledge of the word.
2) Deterministic - actions have guaranteed outcomes.
3) Static - world ~~sees~~ does not change during planning.
4) Discrete - finite states and actions
5) Single agent - no opponents / collaborators.

Components :-

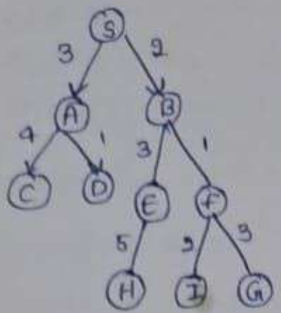
1) Initial state
2) Goal state
3) Actions / operators
4) state space

Examples : Blocks world.

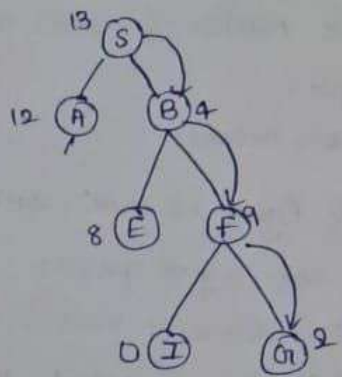
Unit - 2 :-

1. Explain in detail about Greedy best first search.

we are using two lists which are open and closed list. Following are the iteration for traversing the ~~above~~ below example.



node	H(n)
A	12
B	4
C	7
D	3
E	8
F	2
G	4
H	9
I	13
S	0
G	1



Initialization: open[A, B], closed[S]

Iteration 1: open[A], Closed [S, B]

Iteration 2: open[E, F, A], closed[S, B]

Hence final solution path will be : S → B → F → G.

2. Explain crypt arithmetic problem with an example.

- A crypt arithmetic problem is an arithmetic equation where numbers are replaced by letters and you must find the correct digit for each letter.

Rules :-

- Each letter = one unique digit
- No two letters have the same digit.
- First letter cannot be 0.

Example :-

SEND + MORE = MONEY

Solution: 9567 + 1085 = 10652

So: S = 9, E = 5, N = 6, D = 7

M = 1, O = 0, R = 8, Y = 2

∴ this arithmetic means replacing letters with digits to make the equation true.

3. Explain the Minmax Algorithm with an example.

The MinMax Algorithm is used in game playing to find the best move by assuming the opponent also plays optimally.

It is based on two players:

- MAX players.
- MIN players.

→ MAX chooses the move with the highest value

→ MIN chooses the move with the lowest value.

- The algorithm checks all possible moves and picks the best one.

Example :-

```
        MAX
        / \
    MIN     MIN
    / \     / \
   3   5   2   9
```

step 1 : MIN level , tree becomes

```
  MAX
  / \
 3   2
```

step 2 : MAX level, $max(3,2) = 3$.

→ MAX selects value 3 as the best move.

Ex: It is used in AI games like Tic-Tac-Toe , chess..

Unit-3 :-

1. Explain the syntax and semantics of propositional logic with examples.

propositional logic is a way to represent statements and their truth values using symbols.

- It has two main parts:
  - Syntax – structure
  - Semantics – Meaning

1. Syntax of proposition logic:

→ It defines the rules for writing valid logic expressions.

Components:-

1. propositions: It is a statement that is either true or false.
2. Logic connectives: AND, OR, NOT, IMPLIES, IFF.
3. ~~bo~~ Well-formed formulas: valid expressions formed using rules.

2. Semantics of propositional logic:

→ It defines the truth value of expressions.

Ex 1: AND ($\wedge$)

$P \wedge Q$ is true only if both are true.

Example: P: It is raining (T)
Q: It is cold (T)
$P \wedge Q$ = True.

2. OR($\vee$): $P \vee Q$ is true if atleast one is true

3. NOT($\neg$): $\neg P$ means opposite of P

4. Implication ($\rightarrow$)

→ $P \rightarrow Q$ is false only when P is true and Q is false

2. Explain Inference Rules for Quantifiers with an example.

Inference rules for Quantifiers are used in predicate logic to derive new statements from given quantified statements:

1. Universal Instantiation (UI):

Rule: if something is true for all elements, then it is true for a specific element.

Form: $\forall x\, P(x) \Rightarrow P(a)$

2. Existential Instantiation (EI) :-

Rule: if something is true for some element, then it is true for a specific unknown element.

form:

$\exists x P(x) \rightarrow P(a)$

3. Universal Generalization (UG) :-

Rule :- If something is true for any arbitary element, then it is true for all elements.

form: $P(a) \Rightarrow \forall x P(x)$

4. Existential Generalization (EG) :-

Rule :- if something is true for a special element, then it is true for some element.

form :- $P(a) \Rightarrow \exists x P(x)$.

3. Explain Resolution in FOL with an example.

Resolution in FOL is an interface rule used to prove statements true or false by converting them into clauses and resolving contradictions.

- It is mainly used in AI for Automatic theorem proving

Steps in Resolution :-

i) Remove implication ($\rightarrow$)

2) Move NOT($\neg$) inward

3) Standardize variables

4) Convert to clause form

5) Apply resolution rule.

Example :- i) All humans are mortal.

$\forall x$ (Human $(x) \rightarrow$ Mortal(x))

2) Socrates is human

Human (Socrates).

Final Result:

we proved Mortal (Socrates).